



ME 599 J, K / IND E 599 A AI/ML for Smart Manufacturing

Survey, Outlook and Overview

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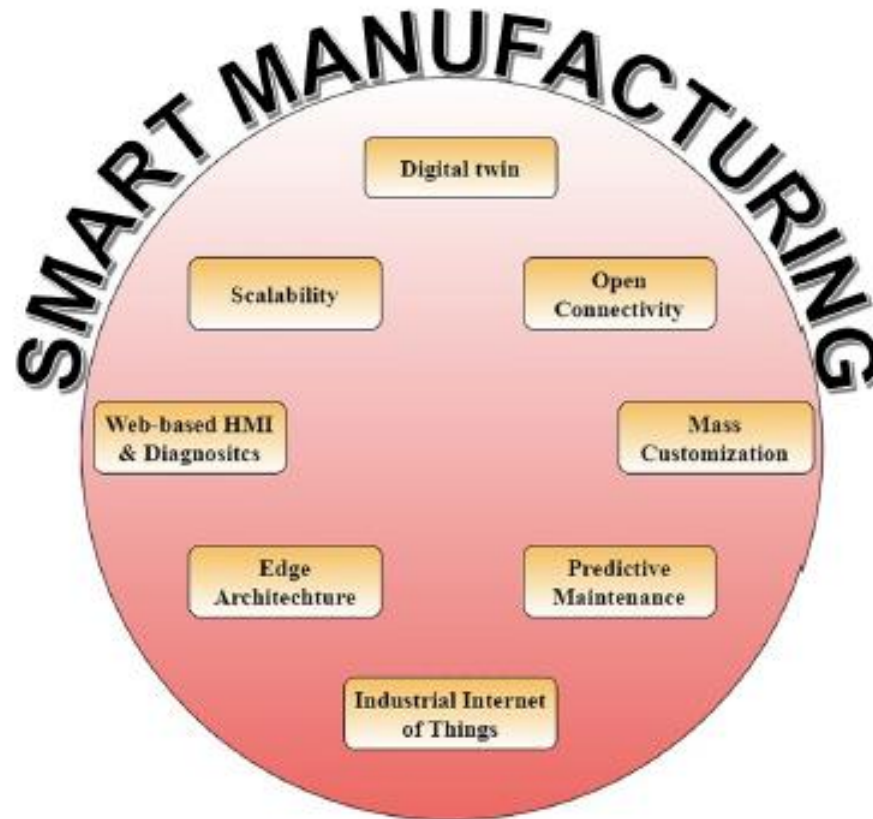
Papers

1. Rai, R., Tiwari, M.K., Ivanov, D. and Dolgui, A., 2021. Machine learning in manufacturing and industry 4.0 applications. *International Journal of Production Research*, 59(16), pp.4773-4778.
2. Dogan, A. and Birant, D., 2021. Machine learning and data mining in manufacturing. *Expert Systems with Applications*, 166, p.114060.



Industry 4.0 $\Rightarrow$ Smart Manufacturing [1]

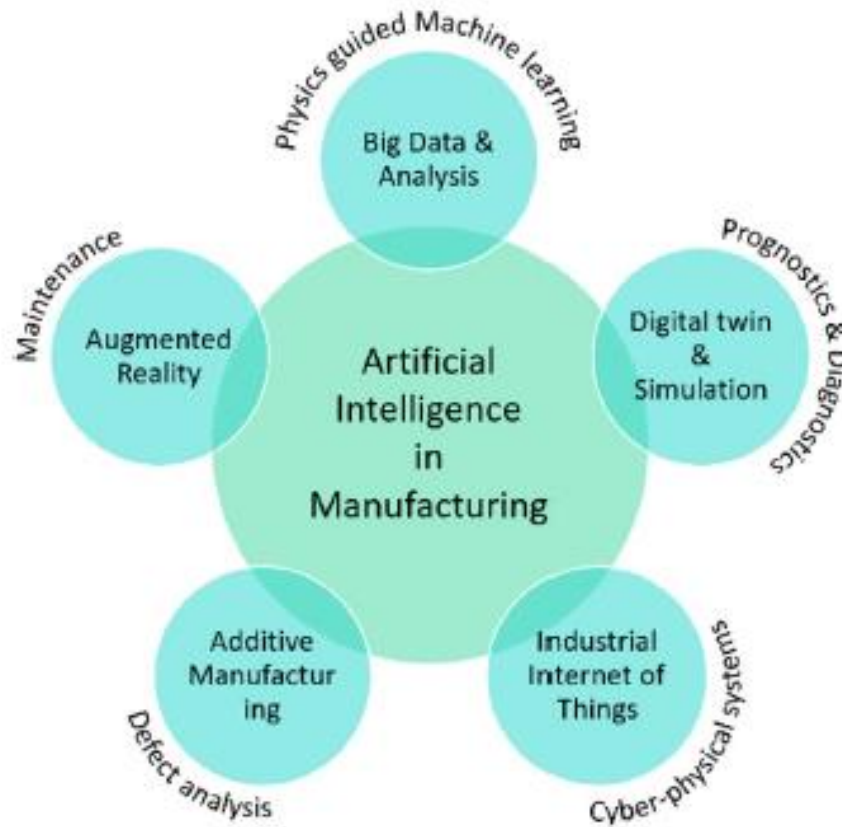
- Industry 4.0 is characterized by
 - Connectivity, intelligence, and flexible automation



What about
cybersecurity and
embodied AI?



Role of AI & ML [1]



Not always “big data” as data acquisition and labeling can be sometimes difficult and expensive

In addition to AM, defect analysis can be critical in other challenging and important processes such as nanofabrication, automated fiber placement, etc.



Applications [1]

- Improved assembly line efficiency
- Improved customer experience
- Better inventory management
 - Real-time visibility
 - Route delivery optimization
- Reduced errors and waste during product development
- Predictive maintenance
- Real-time asset management to minimize losses
- Increased visibility into supply chains
- What about parts designs selection and materials discovery/selection?



Focus Areas [1]

- Computer vision-based inspection and monitoring
 - Automated inspection of IC wire bonding defects [3]
 - Soft sensor for floatation froth grade classification [4]
- Fault detection
 - Fault diagnosis of rolling bearing [5]
 - Fault detection of industrial cold forging [6]
- Process improvement and optimization
 - Utilization of different data sources for injection molding [7]
 - Process mining in make-to-stock manufacturing [8]
 - Optimized assembly line feeding mode selection [9]
- Cloud manufacturing
 - Edge-based hybrid scheduling learning model [10]



Conclusions [1]

- Manufacturing is undergoing a fundamental paradigm shift
- Manufacturing environments are turning into cyber-physical systems where AI is indispensable
 - Additional applications include geometric processing, material informatics, and control of dynamic systems
- Synergy of ML and physics is becoming crucial
 - Data generation can be difficult and expensive
- Anything not discussed?



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8. Lorenz, R., Senoner, J., Sihn, W. and Netland, T., 2021. Using process mining to improve productivity in make-to-stock manufacturing. *International Journal of Production Research*, 59(16), pp.4869-4880.
9. Zangaro, F., Minner, S. and Battini, D., 2021. A supervised machine learning approach for the optimisation of the assembly line feeding mode selection. *International Journal of Production Research*, 59(16), pp.4881-4902.
10. Jian, C., Ping, J. and Zhang, M., 2021. A cloud edge-based two-level hybrid scheduling learning model in cloud manufacturing. *International Journal of Production Research*, 59(16), pp.4836-4850.